Artificial sources are man – made light sources such as flame of candle, incandescent lamp, neon lamp, Sodium lamp etc.

1. Incandescent Sources: When certain objects are heated to a high temperature, they begin to emit light. The glowing of hot iron rod is a kind of Incandescent light.

Example: Candle, incandescent lamp.

2. Gas Discharge Sources: Passing electricity through certain gases at a very low pressure (discharging) can produce light .

Example: Neon lamp, Sodium lamp



We often use a kind of gas-discharge lamp that uses fluorescence to produce visible light. The electric current in the gas excites mercury vapour, which produces short-wave ultraviolet light that then causes a phosphor coating on the inside of the lamp to glow in visible light.



Properties of light

In this section, we shall examine some properties of light. Light has some fundamental properties as mentioned below

- ❖ Rectilinear propagation of light
- ❖ Reflection
- ❖ Speed
- ❖ Interaction of light with matter
  - Types of material according to permeability
  - Formation of shadows
  - Plane mirror and images
- ❖ Spectrum

The path of light

How does light travel?

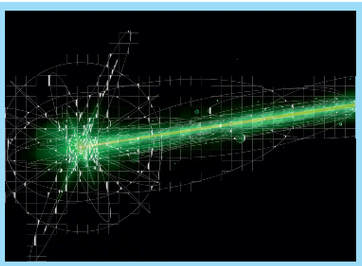
- Have you ever seen the scene of light penetrating through the branches of trees in denser forest?
- Have you ever seen the path of sun light entering through the hole of a cement grill building?
- Have you ever seen the path of a laser light?



Light penetration through trees



Light travels through the holes in cement grill



Laser beam

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